Nick MAHTANI

Proactive | Passionate | Creative

+41 78 966 22 00 • nikhilmahtani@outlook.com

RESEARCH EXPERIENCE

Aug 2025 – Present **Self-Directed Computational Biology & Machine Learning Training**

- Reproducing single-cell RNA-seq analysis pipelines from IHB publications (He et al., Nature 2024; Jain et al., Nature 2025)
- Completing coursework: Machine Learning (CS-433, Bob West, EPFL), Applied Computational Genomics (Aaron Quinlan, Utah), Machine Learning for Computational Biology (Manolis Kellis, MIT)

Jan 2024 – Jun 2025 **Laboratory of Chemical Nanotechnology (CHEMINA), EPFL, Switzerland**

Scientific Assistant, Biosensors for Small Molecule Metabolites in Complex Biological Fluids

- Developed a method to integrate DNA aptamers with laser-based focal molography for serotonin detection in human serum
- Quantified thermodynamics of aptamer interactions with serotonin and dopamine using isothermal titration calorimetry
- Designed and analyzed DNA aptamer sequences using NUPACK
- Characterized aptamer-modified nanopipettes with TEM imaging
- Supervised two students (summer internship and master's thesis)
- Presented work at the Maxwell Biosystems Summit 2024 and the 2024 Symposium on Parkinson's Disease and Related Disorders

Sep 2022 – Apr 2023 **Mooney Lab of Cell and Tissue Engineering, Harvard University, USA**

Visiting Master's Thesis Student, Macroporous Alginate Scaffolds for Tendon Tissue Engineering

- Synthesized macroporous alginate scaffolds with tunable pore size and formation kinetics
- Generated fluorescently labeled alginate (FITC) using carbodiimide chemistry
- Visualized pore formation in fluorescently labeled alginate gels using confocal microscopy
- Quantified primary tendon cell proliferation and release as a function of pore size using EdU assays
- Developed a novel method to induce controlled porosity using pressurized airflow through a nebulizer
- Purified oxidized alginate samples via tangential flow filtration
- Presented work at ORS 2024

Aug 2021 – Jan 2022 **Lütolf Lab of Stem Cell Bioengineering, EPFL, Switzerland**

Research Assistant, Liver Organoid-on-Chip Technologies

- Established and maintained liver organoid-on-chip microfluidic cultures using primary human hepatocytes and non-parenchymal liver cells
- Optimized organoid viability by tuning culture conditions like cell ratio, cell types, and ECM composition
- Designed and conducted experiments to induce vasculature by creating VEGF gradients across the chip
- Characterized organoid models using confocal microscopy and immunofluorescence staining for relevant markers
- Successfully generated and maintained vasculature for up to two weeks in culture

Jul 2021 – Aug 2021 **Aztekin Lab of Structural Regeneration, EPFL, Switzerland**

Research Assistant, Development of an In Vitro System to Investigate AER Cells

- Designed, cloned, and transfected plasmid constructs to overexpress transcription factors in mESCs
- Applied perturbations to stem cells using biologically relevant signaling molecules to direct differentiation
- Performed qPCR experiments to track gene expression profiles and analyzed data obtained
- Implemented HCR and immunofluorescence staining protocols
- Generated organoid and gastruloid co-cultures

Feb 2021 – Jul 2021 **Renaud Lab, Microsystems Laboratory 4, EPFL, Switzerland**

Research Assistant, Droplet-Based Single-Cell Encapsulation for Cancer T Cell Therapy

- Employed droplet microfluidic technologies for single-cell encapsulation
- Optimized chip design to increase sorting efficiency of droplets encapsulating single cells
- Performed data analysis and mapping

Sep 2019 – Aug 2020 **Lab for Immuno Bioengineering Research and Applications, NYU Abu Dhabi**

Research Assistant, High Throughput Bioprinting of Immune Cell-Laden Scaffolds

- Transformed a Prusa 3D printer into a bioprinter capable of extruding cell-laden bioinks
- Fabricated complex 3D collagen-based scaffolds containing immune cells
- Induced varying degrees of collagen fiber alignment and observed effect on antigen-presenting immune cells

INDUSTRY EXPERIENCE

Mar 2022 – Sep 2022 **CSEM (Swiss Center for Electronics and Microtechnology), Neuchâtel**
Intern, Tools for Life Sciences Unit, Liver Tissue Engineering

- Fabricated perfusable collagen hydrogels for liver organoids
- Designed and tested prototypes for microfluidic smart lids for pooling multi-well plates
- Developed components to enable passive continuous perfusion for on-chip tissue cultures
- Optimized functioning of perfusion bioreactor for liver tissue engineering

EDUCATION

Sep 2020 – Apr 2023 **École Polytechnique Fédérale de Lausanne (EPFL), Switzerland**
Master of Life Sciences Engineering (Specialization: Biomedical Engineering) **GPA: 5.57/6**

Jul 2021 **Utrecht University, Netherlands**
Summer School, 3D Printing and Biofabrication

Sep 2015 – Jun 2019 **American University of Sharjah (AUS), Sharjah, UAE**
Bachelor of Science in Mechanical Engineering **GPA: 3.7/4 – Magna Cum Laude**

Apr 2000 – Mar 2015 **Delhi Private School, Sharjah, UAE**
Results: 93.4/100 **Computer Programming School Topper: 99/100**

PUBLICATIONS

Sapudom, J., Karaman, S., Quartey, B. C., Mohamed, W. K. E., Mahtani, N., et al. (2023). Collagen Fibril Orientation Instructs Fibroblast Differentiation Via Cell Contractility. *Advanced Science*, 10(22).

Sapudom, J., Mohamed, W. K. E., Garcia-Sabaté, A., Alatoom, A., Karaman, S., Mahtani, N., & Teo, J. C. M. (2020). Collagen Fibril Density Modulates Macrophage Activation and Cellular Functions during Tissue Repair. *Bioengineering*, 7(2), 33.

ACHIEVEMENTS

- Awarded Zeno Karl Schindler scholarship to carry out master's thesis at Harvard University
- Represented EPFL in the SensUs Student Competition 2021 as part of Team SenSwiss: 1st place Analytical Performance Award and 2nd place Translation Potential Award
- Represented AUS as a delegate at the 22nd World Business Dialogue in Cologne, Germany (2019)
- Placed on the Chancellor's Distinguished List for three years
- Received the Petrofac Endowed Scholarship and Abu Dhabi Islamic Bank Scholarship (2018)

TECHNICAL SKILLS

Programming: Python, R, Bash, C++, JavaScript, MATLAB
Data Analysis & Visualization: GraphPad Prism, ImageJ
Molecular Biology Tools: SnapGene, NUPACK
CAD & Simulation: SolidWorks, AutoCAD, Autodesk Inventor, Fusion 360, COMSOL, Ansys Fluent

EXTRACURRICULAR ACTIVITIES

- Co-founded The Brain Hub Podcast (2024): a platform for early-career researchers in medicine and biomedical engineering to share their research journeys
- Created Cell Stories YouTube channel (2022–2023): videos explaining complex research topics in simplified, engaging formats
- Involved in musical theatre, choir, and production pieces at AUS (2016–2019)
- Volunteering trip to teach at Myuni School in Zanzibar, Tanzania (2019)
- Member of AUS Track and Field team (2015–2016)